



License Plate Recognition Systems

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INTRODUCTION

License Plate Recognition (LPR) Systems are a surveillance method that use optical character recognition (OCR) on video images to read license plates on motor vehicles. LPR camera systems can be mounted on police cars or in fixed locations. Typically, the cameras are outfitted with software that searches for the presence of a license plate. Once one is detected, the image is captured and then OCR extracts the letters and numbers on the license plate. The extracted data can then be stored, linked to other applications, or compared to information in databases. LPR systems are comparable to [face recognition](#) systems, which are computer-based security systems that can automatically detect and identify human faces.

LPR, known as automatic number plate recognition (ANPR) in the United Kingdom, originated in the UK where it is extensively used to track UK vehicle movements in real time. Collected data is stored for five years in the National ANPR Data Centre, and may be analyzed for intelligence and used as evidence.

LPR Systems Can be Used in a Number of Ways

- For police enforcement
- As average speed cameras
- For electronic toll collection
- To monitor border crossings
- At gasoline filling stations

- For traffic management
- For drive through customer recognition

Examples of how LPR systems are used in law enforcement include stolen vehicle identification, identifying wanted felons, sexual predator and DUI surveillance, drug enforcement, general surveillance and investigation, and homeland security purposes. Law enforcement agencies across the nation have or plan to acquire LPR systems. Departments in Colorado, Maine, Georgia, Connecticut, Washington DC, New York, Michigan, Virginia, Florida, New Jersey, Tennessee, California, Pennsylvania and Kentucky are currently known to use LPR cameras.

Although the number of license plates that can be scanned differs depending on the technology adopted, LPR systems can quickly collect and process large volumes of license plates. For example, an LPR system in Nassau County, New York, can scan 8,000-10,000 license plates during an officer's eight-hour shift.¹ The Arcadia Police Department in California has stationary license plate scanner cameras that can scan 1,500 plates per hour.² In Philadelphia, police use a van equipped with LRP cameras to roam the streets scanning around 3,600 plates per day.³

LPR Systems and Privacy

EPIC highlighted the dangers of location-tracking technology in People, Not Places, A Policy Framework for Analyzing Location Privacy Issues.

Similar to the federal Privacy Act of 1974, some states (including California, Hawaii, Idaho, Kentucky, Massachusetts, Minnesota, New York, Ohio and Virginia) have adopted equivalent laws that may affect how LPR systems can be used by law enforcement agencies. While there are variations among the state privacy acts, there are some basic commonalities such as the requirement that state agencies collecting personal information must be authorized by law to collect such information, or that collection of such information be necessary for the agency to perform its duties. State agencies are required to provide notice to individuals, informing them of the collection of personal information and the purpose for which such information is collected. There are also often administrative procedures whereby individuals may demand that personal information about them may be corrected or deleted. There should be a remedy ensuring compliance with the act. Lastly, state agencies are required to maintain records with reasonable accuracy, relevance, timeliness and completeness. Use of LPR systems should be subject to state privacy act requirements of notice, access and correction.

The use of LPR systems and collection of license plate data implicates informational privacy interests. Informational privacy is concerned with the collection and dissemination of data, technology and the public expectation of privacy. Although U.S. courts have not recognized a reasonable privacy expectation in a license plate⁴, the collection and use of license plate data through LPR systems raise concern for informational privacy. While the information on a license plate itself may not be private, the use of such collected information raises privacy concerns. Courts have found that drivers have a right to be free from warrantless GPS tracking. In Commonwealth v. Connolly, the Massachusetts Supreme Judicial Court barred warrantless GPS tracking on public streets, recognizing citizens' interests in keeping their travels private. Other courts have held similarly. Some location privacy cases remain pending.

Furthermore, EPIC has documented that government databases, such as the Employee Eligibility Verification database and the FBI's National Crime Information Center database, are filled with errors. For example, in Herring v. US, police searched and then arrested Bennie Dean Herring based on incorrect information in a government database. Creating and maintaining databases with license plate location information may have potentially serious privacy implications for Americans if such databases are not subject to conditions or are abused. For example, the Maine Legislature's Joint Transportation Committee moved an amended bill (LD 1561) forward that would set statewide policy for the use of LPR systems. Compromise was reached where such systems can only be used when police are performing a criminal investigation, are worried about public safety, are enforcing civil orders, or are responding to law enforcement bulletins.

After 21 days, the collected data should be purged. Other than Maine, New Hampshire is the only other state to restrict the use of LPR systems. In 2007, New Hampshire passed SB41, which bans the use of any surveillance technologies on the public, including LPR cameras.

Questions About LPR Systems

- Does the LPR system monitor all motor vehicles and retain license plate identification information on all citizens?
- Is information collected by the LPR system saved? Is it retained? For how what duration?
- Does the system include controls over who has access to license plate information?
- Are binding laws in place, rather than departmental policies, governing how such information may be used?

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